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AI ETHICS FRAMEWORK: BALANCING AUTOMATION AND CITIZEN PROTECTION

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Addressing Power Imbalance Between Citizens and Institutions

Project Overview

This report is about the Dutch Toeslagenaffaire (childcare benefits scandal) which wrongfully accused over 26,000 parents of welfare fraud. The tax authority used discriminatory algorithms and harsh "all-or-nothing" policies. Victims lost homes, suffered severe debts, and had thousands of children unjustly taken into foster care, ultimately forcing the government's resignation. We are tasked with rebuilding trust after the Dutch childcare benefits scandal.

Mission

Design an AI system for welfare allocation that prioritizes ethics, compliance, and human dignity from the outset.

Problem definition

How can we prevent algorithmic harm while ensuring efficient service delivery.

The purpose of this report is to bridge the technical foundations of algorithmic fairness with its wider socio-technical context, where the link between technical accuracy and real-world stakeholder impact becomes clear. By prioritizing fairness towards stakeholder outcomes, the approach aims to prevent disproportionate harm to individuals or groups such as the injustices witnessed during the Dutch Toeslagenaffaire. Because these impacts are measurable, improvements can be directed toward the areas that genuinely cause harm.

1. Ethical Autopsy

1.1 Floridi's AI Ethics Framework

Evaluating past mistakes is essential for continuous improvement. By learning from previous ethical failures in AI, particularly those lacking a human touch. We can establish concrete regulations and safeguards to prevent history from repeating itself. To effectively visualize and analyze these past missteps, we will apply the Floridi AI Ethics Framework.

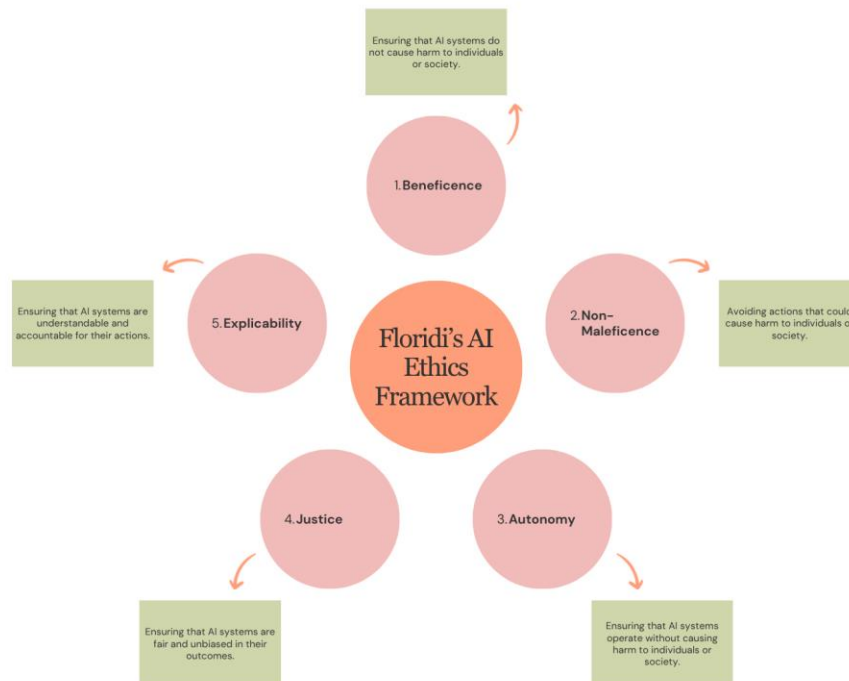


Figure 1. Floridi's AI Ethics Framework.

Beneficence

In this case, there was a lack of preservation of human dignity where 26,000 parents of welfare fraud were falsely accused. Many lost their homes, were evicted and became

homeless due to harsh all or nothing policies. They criminalized innocent people, suffered severe debts, and had thousands of children unjustly taken into foster care. This harmed instead of contributing to the common good and sustainability. AI was not being used for the wellbeing but did quite the opposite.

Non-Maleficence

According to this principle, AI systems must do no harm. The agency used unregulated algorithms to flag "high-risk" claims. The system was heavily biased and illegally targeted low-income families and people with dual or foreign nationalities.

Autonomy

Human agency and oversight were entirely absent. Officials did not bother to question the algorithm, automatically approving its choices, and the process was too automated, turning humans into mere executioners of flawed digital outputs rather than thoughtful decision makers.

Justice

AI must promote fairness. This system was heavily biased and illegally targeted low-income families and people with dual or foreign nationalities. A parliamentary inquiry later officially recognized that ethnic profiling and institutional racism drove the agency to aggressively target certain minority groups for minor bureaucratic mistakes actively denying basic civil protections.

Explicability

The system severely lacked both intelligibility and accountability. It was used as a "black box," and parents or oversight bodies could not understand how the algorithm reached its decisions, nor did officials clearly take responsibility.

1.2 Harm Assessment

It separates the risks into two categories: immediate damage to individuals and broader negative impacts on society.

This matrix assesses the harms caused by the AI-driven harassment and abuse in this case.

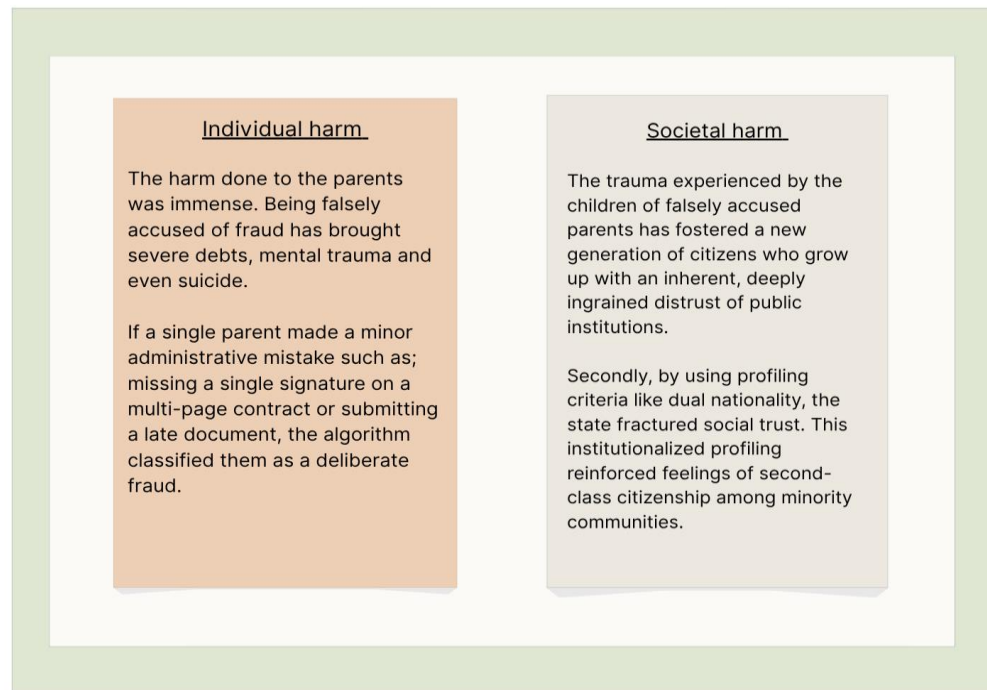


Figure 2. Harm Assessment.

Reflection

At first sight the cause seemed to be the model, because that's where the manipulation of the data occurs, Afterwards, it became evident that the dataset contained columns such as dual nationality and low income attributes that should have been excluded or masked, as their use represents an unacceptable risk and is prohibited under the EU AI Act. If an "ethics-by-design" framework had been applied, the data phase would have most fundamentally changed the outcome of the *toeslagenaffaire*.

While failures occurred across all stages, the model and deployment phases merely amplified and automated the biases that were institutionalized right at the beginning. Fixing the data phase would have stopped the scandal at its very source.

2. Governance Blueprint

2.1 Roles & Accountability

To eliminate blind spots, institutional silos, and automation bias, this blueprint establishes an independent oversight chain with explicit escalation mandates. The AI Governance Matrix in figure 3. provides a structured overview of accountability roles, oversight mechanisms, and escalation pathways within algorithmic decision-making systems. It clarifies how responsibilities are distributed from data ownership and model validation to ethical compliance and citizen recourse ensuring transparency, fairness, and adherence to the EU AI Act.

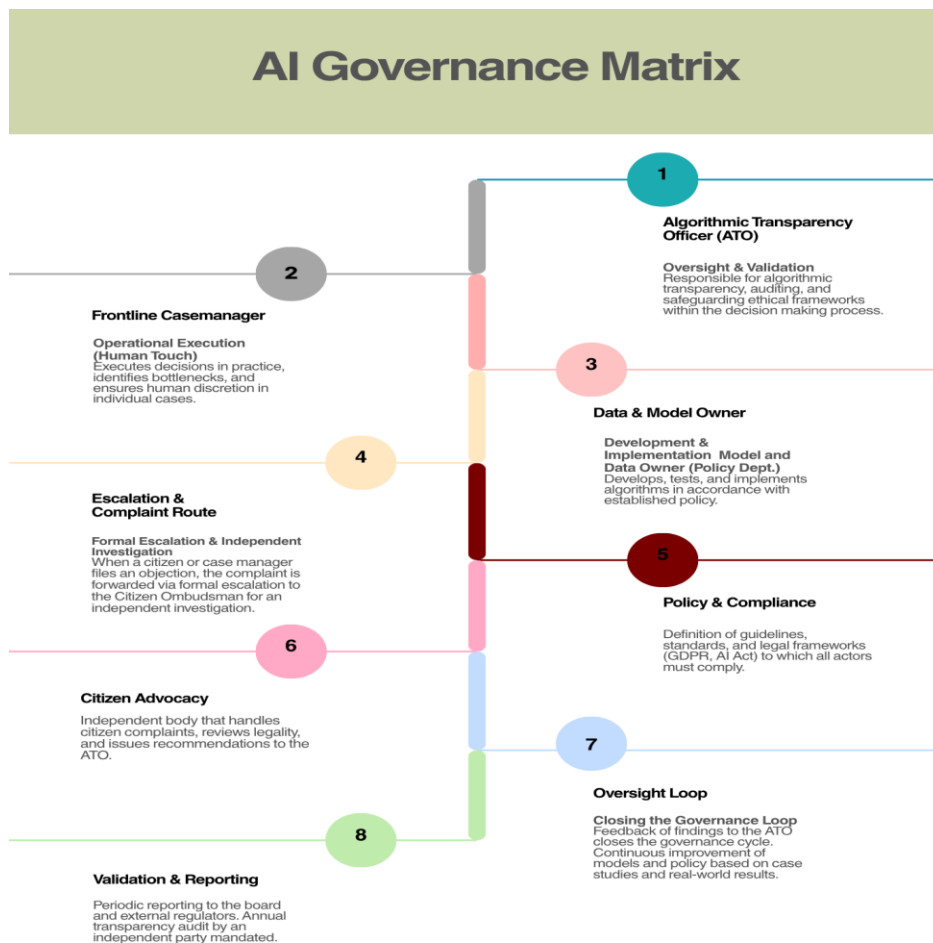


Figure 3. AI Governance Matrix- Overview accountability roles and oversight mechanisms.

To eliminate blind spots, institutional silos, and automation bias, this blueprint establishes an independent oversight chain with explicit escalation mandates.

Task/Role	Core Task	Responsibility	Mandate
Algorithmic Transparency Officer	Conducts and enforces Fundamental Rights Impact Assessments (FRIA/IAMA).	Ensures EU AI Act compliance, bans sensitive proxy data, and manages public register.	Absolute veto over system deployment; reports to political leaders.
Data Model Owner	Monitors system for model drift and demographic parity.	Prevents specific groups from being disproportionately flagged.	Legally accountable for real world performance, safety, and ethics.
Frontline Casemanager	Manually verifies every "high-risk" flag before any negative action.	Provides clear, written justifications to citizens selected for review.	Autonomous power to overrule algorithmic recommendations without sanction.
Citizen Ombudsman	Acts as a direct escalation point for wrongfully targeted citizens.	Enforces swift manual re-evaluations and reports systemic issues to Parliament.	Independent oversight body outside the agency's hierarchy.

Figure 4. Table Tasks & Roles of Blueprint.

2.2. Legal Anchors

GDPR Article 22 Constraining the Design

GDPR Article 22 establishes that individuals have the right *not* to be subject to a decision based solely on automated processing if it produces legal or significantly disruptive effects. The *toeslagenaffaire* bypassed this via a blind approval culture where civil servants blindly trusted automated fraud scores. This blueprint enforces meaningful human intervention through two strict architectural constraints and one design principle:

No Automated Execution Enforcement: The algorithm has zero technical read or write access to the payment infrastructure. It cannot pause, reduce, or demand repayments of benefits. It is strictly constrained to generating a ‘recommendation for review’.

Mandatory Analytical Justification: The system's interface forces case managers to actively analyze the original data. The software blocks the user from proceeding or closing a case until they manually type out a qualitative justification explaining why they agree or disagree with the algorithmic signal.

Right to Explanation by Design: The system automatically outputs a clear, simplified "Explanation Card" based on local feature importance such as SHAP values. This document is dispatched directly to the citizen, showing exactly which factors triggered the review, ensuring they can exercise their right to a swift and informed appeal.

Classification system under the EU AI Act

Although the Dutch Toeslagenaffaire showed harms comparable to “unacceptable risk”, under the EU AI Act it falls within Annex III Sections 5 and 6 as a high-risk system, because welfare eligibility algorithms are regulated rather than prohibited. The algorithm is implemented by a public authority to assess citizens’ eligibility for social security benefits, such as childcare allowances. and to detect and prevent financial misconduct (fraud). Mistakes in this domain directly compromise citizens' fundamental rights to financial security, non-discrimination, and family life.

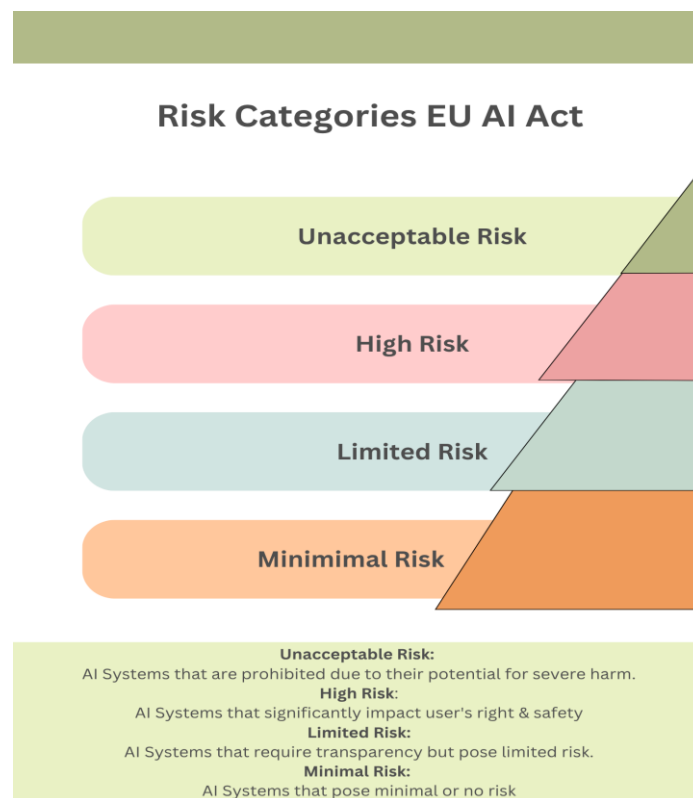


Figure 5. Visual Risk Categories EU AI Act.

Design Obligations:

This classification legally binds the agency to perform a pre-market Fundamental Rights Impact Assessment, register the system in the public EU AI Database, institute a rigorous Quality Management System (QMS), and implement continuous, untamperable logging to ensure total transparency throughout the system's operational lifecycle.

2.3. Bias Mitigation

Technical safeguard: Bias Labeling and Removal

The most effective technical safeguard in the context of the *Toeslagenaffaire* lies at the data preparation stage. The harm originated before modeling in the features chosen, and the labels applied (fraud vs. non-fraud). Columns such as nationality and dual citizenship acted as proxies for ethnicity, embedding systemic bias directly into the training data. Once those features existed, even a fairness aware model would still reproduce discrimination, because it was learning from biased examples.

Bias Labeling and Removal prevents discriminatory features such as nationality and income proxies from entering the model. By identifying and excluding sensitive or proxy attributes before training, it ensures that bias is mitigated at its source. This process includes:

- Flagging protected attributes such as nationality, income, or family structure.
- Testing correlations between these attributes and target labels.
- Removing or neutralizing them through feature de-correlation or re-weighting.

To strengthen this safeguard, Disaggregated Error Analysis should be applied during evaluation to detect any residual bias that slips through. Together, these measures create a robust defense against systemic discrimination by addressing bias both before and after model training.

Procedural Safeguard: Community Review Board

Algorithms cannot understand historical context, social inequality, or systemic racism. Purely statistical checks are blind to societal impact.

- **The Mechanism:** An independent, diverse advisory board is established to oversee the algorithm. It is composed of data ethics experts, civil rights lawyers, representatives from public advocate offices such as the National Ombudsman), and citizens who rely directly on government social welfare programs.
- **Application:** Before any model training or system updates occur, this board must review and approve the exact variables utilized and the definition of "fraud." They hold the legal authority to flag operational patterns where automated monitoring is resulting in the social exclusion or stigmatization of specific neighborhoods or communities.

The Community Review Board is the most logical procedural complement to Bias Labeling and Removal. While the technical safeguard detects and removes bias within data, the board ensures that data choices and definitions are ethically justified and socially legitimate, creating a continuous oversight loop that prevents systemic discrimination.

3. Redesign for Trust with Human-Centered Solutions

3.1. Explainability Decision Making

To ensure an administrative disaster like the *Toeslagenaffaire* never happens again, the automated system must be redesigned with transparency, human control, and accessible justice at its core.

To maintain clear explainability and uphold transparency, it is essential that every automated decision can be easily understood by non-technical users. Decisions should be communicated in plain language, without requiring expertise in computer science or tax law. Below are several practical solutions that can be integrated to make algorithmic decisions more accessible and comprehensible to the public.

- **Plain-Language Automated Notices:** Outbound letters will completely ban dense legal jargon and obscure system error codes. If a benefit is adjusted, the notice will lead with a direct summary statement: *"Your childcare benefit was lowered by €120 this month because our system shows 6 hours less daycare usage than reported."*
- **The "Why Button" Visual Dashboard:** Parents can log into their secure personal portal, such as DigiD and see a visual breakdown of their decision tree.
- **Comparative Visuals:** Side by side bar charts will show Reported Data vs. System Verifications like matching the parent's inputted hours directly against the daycare center's submitted data, highlighting the exact discrepancy in red.
- **Jargon-Free Tooltips:** Hovering over any term for example "Income Threshold" instantly triggers a one-sentence pop-up explanation in plain language.

3.2. Human Oversight Protocol

An algorithm should never possess the autonomous power to stop, freeze, or retract a citizen's financial livelihood. High stakes decisions require strict human oversight. Low stakes cases can be fully automated. For instance minor administrative changes that result in increased benefits or trivial adjustments under €50 are processed automatically. High stakes cases have a mandatory human oversight. Any algorithmic output that proposes a benefit suspension, a fraud label, or a repayment recovery exceeding €500 automatically freezes. The AI is blocked and flagged from executing the action.

The system routes the flagged case to a human caseworker's queue. The caseworker must complete a three-step checklist before the system allows them to hit "Submit".

- **Source Check:** Did the AI flag a malicious act, or just a missing signature/clerical typo?
- **Proportionality Check:** Is the proposed penalty proportional to the error, or will it cause immediate poverty?
- **Contact Rule:** The caseworker must attempt to call or email the citizen to resolve the administrative gap, before pushing any penalty forward.

The flowchart below provides a clear visualization of the full process, showing how automation operates and when human oversight takes effect.

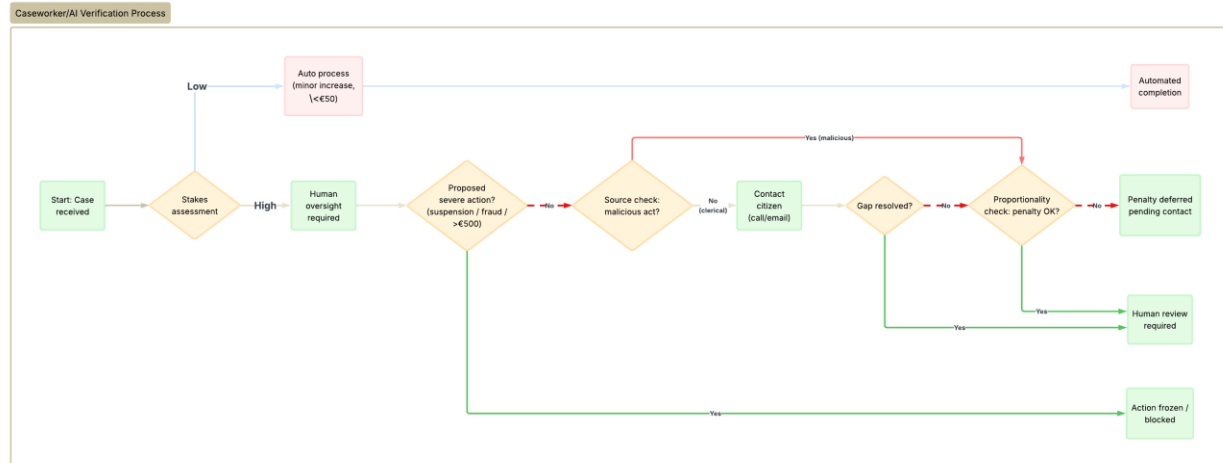


Figure 6. Caseworker/AI Verification Process.

3.3. Redress Mechanism

If a citizen disagrees with a decision, they must have access to a swift, compassionate, and transparent path to rectification. This two-step citizen appeal process enables the redress mechanism to function effectively.

Step 1: Immediate Automated Review

- **The Action:** Within their portal, the user clicks a prominent "Request Review" button next to the disputed decision.
- **The Process:** The user is guided through a simple form where they can state the error and directly upload supporting evidence for example a missing receipt or a corrected bank statement.
- **The Logic:** The system runs a targeted, single-variable check on the newly uploaded data. If it clears the inconsistency, the system instantly reverses the penalty, updates the benefit, and issues an apology notice within 48 hours.

Step 2: Formal Human Caseworker Appeal

- **The Escalation:** If the automated review fails, or if the user is dissatisfied with the automated outcome, the case is automatically escalated to Step 2.
- **The Process:** A dedicated human caseworker which is completely independent from the original team that handled the initial flag, is assigned to the file.
- **The Mandate:** The caseworker is legally required to review the case using the "Principle of Proportionality." They must look at the human context rather than strict code.
- **The Resolution:** The caseworker schedules a phone call or in-person meeting with the citizen to discuss the case. A final, legally binding human decision must be delivered within 30 days, heavily prioritizing citizen protection over rigid system compliance.

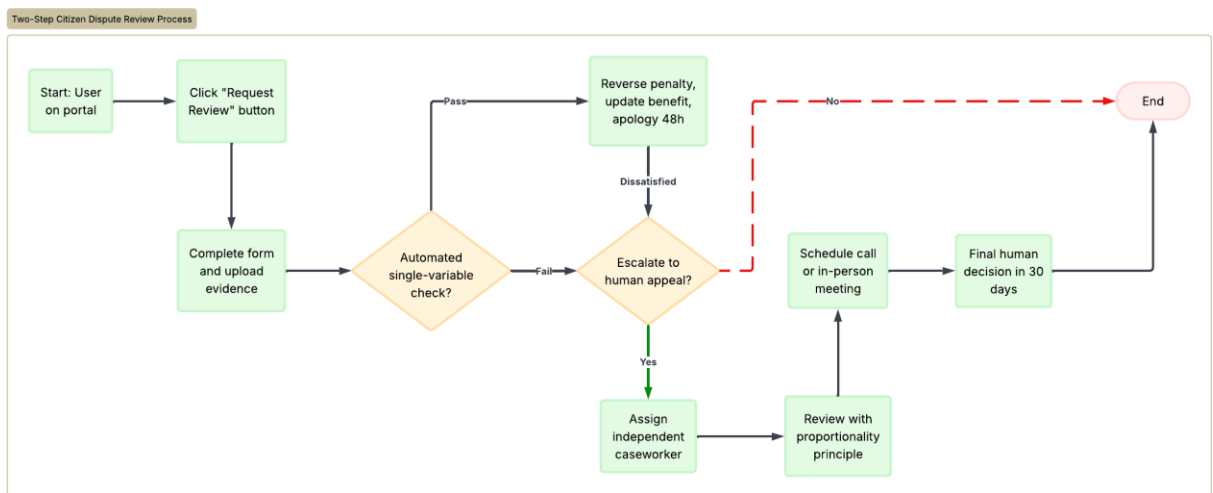


Figure 7. Two-Step Citizen Dispute Review Process

Reflection

This design addresses the power imbalance between citizens and institutions by introducing stronger human oversight and using AI in a smarter, more efficient way. It ensures that automated decisions are transparent, reviewable, and always subject to human judgment when stakes are high. By combining explainable AI tools with clear citizen review stages, the system restores trust and accountability in digital governance.